Inferring Business Processes from Desktop Operation Logs

A Literature, Open-Source and Industry Survey for the
"From Operation Logs to an Automation Proposal" Assignment

Prepared by
Indrani Chattopadhyay

Assignment
Forward Deployed Engineer selection task
Duration: 7 days · Deliverables: `segments.jsonl`, repository, report, work log

IIT BHUBANESWAR

Survey revision: v2 (two research passes)
Date: 10 September 2026

This document is a *fact-finding survey*. It deliberately proposes no architecture, method, automation target, technology stack or project plan. Its purpose is to establish what prior work exists, what that work assumes, where its published results stop, and which claims about this assignment would therefore require careful evidence. All figures are original diagrams prepared for this survey; none reproduces a figure from a source paper.

Abstract

A client company records employees' desktop operations — keystrokes, clicks, application switches, clipboard changes, window titles, browser interactions, occasional extracted screen text and selected screenshots — while they perform back-office work. The logs carry no business-process labels, no case identifiers and no task boundaries. The assignment surveyed here asks for three connected things: recover coherent units of work from those streams and label repeated executions consistently; analyse a second, unlabelled production dataset drawn from *different* departments and applications to identify and rank automation opportunities; and build a working prototype, defending its scope, residual manual work and rollout risks. This survey establishes what the literature, open-source ecosystem and commercial market already offer for that problem. Two research passes found no benchmark, competition or public dataset matching the assignment, and no end-to-end published method operating under its constraints. Seven fields cover its components separately: robotic process mining, event abstraction and case-identifier discovery, desktop activity recognition, linear segmentation evaluation, trace clustering, multimodal user-interface understanding, and RPA candidate selection. The survey's central finding is that the field's flagship method for discovering routines from unsegmented interaction logs states in its own limitations that it requires consecutive, non-overlapping routine executions that do not share actions, and that it degraded on the one real log characterised by frequent application switching — precisely the three conditions this assignment's data is described as having. A 2021 review of segmentation independently reports that no published work handles actions shared across concurrent routine executions. Meanwhile the best-documented commercial products require a human to define the task before analysis begins, and the products claiming fully automatic discovery publish no verifiable method. Sixteen evidence-linked limitations are distilled, together with the claims about this assignment's data that an informed evaluator could most easily challenge, and fourteen questions that only the datasets themselves can settle.

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1 Introduction

Back-office work in HR, finance and logistics departments is largely invisible to the systems that support it. An expense claim leaves a trace in the finance system; the twenty minutes a clerk spent moving figures between a portal, a spreadsheet and a letter template leaves no trace anywhere except on the clerk's own screen. Desktop telemetry agents were built to close that gap, and they succeed at the mechanical level: they record every keystroke, click, application switch, clipboard change and window title. What they do not record is *meaning*. A log says that a user pressed eleven keys in Excel, switched to Chrome and copied a string. It does not say that an expense claim was being processed, that this was the fourth such claim that morning, or that the claim was suspended halfway through when a colleague asked a question.

The assignment that motivates this survey asks an analyst to close that second gap and then act on it. Concretely: recover units of work from an unlabelled operation-log stream and label repeated executions of the same process consistently; apply that recovery to a production dataset from departments and applications never seen during development; quantify what work is performed, how often, by how many people and at what relative cost; rank automation opportunities; and build a prototype, justifying scope, residual manual effort and rollout risk. The measurable dataset and the deliverable dataset are deliberately different, and seven days are allotted for all of it.

The purpose of this document is narrower than a general review of process mining. It is to establish, from the published record, four things:

(i) whether this problem — or a close variant — has already been posed as a benchmark, competition, dataset or method, so that the assignment can be positioned rather than guessed at;

(ii) which of its components are mature research areas with usable results, and which are acknowledged open problems;

(iii) what prior work *assumes* about its input, since the assignment withholds nearly every one of those assumptions simultaneously; and

(iv) which claims about the assignment's data would, in the light of that record, require unusually careful evidence.

Figure 1 situates the assignment against the pipeline that the closest field — robotic process mining — already defines for exactly this problem, and marks the two stages that field describes as unsolved.

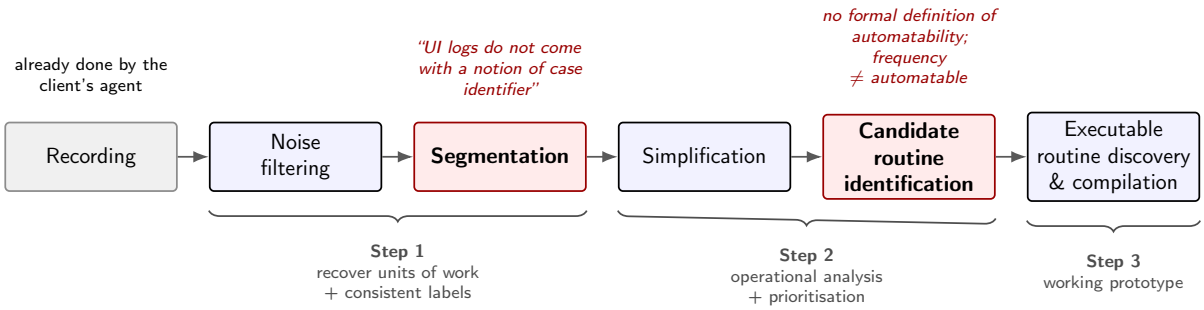


Figure 1: The six-stage robotic process mining pipeline mapped onto the three steps of this assignment. Original diagram prepared for this survey; the stage sequence and the two red annotations follow the pipeline definition and the stated open problems in Leno et al. [31]. Recording is already done by the client's agent, so the assignment begins at stage two — and the two stages it depends on most are the two the field describes as unsolved.

The survey deliberately stops short of recommending anything. It reports what exists, what it assumes, what it measured, and where it stops. Section 9 converts that into evidence-linked limitations, and Section 10 lists the questions that no amount of reading can answer because

only the datasets can.

2 Scope and Terminology

Much of the confusion in this area comes from using one word — "task", "process", "activity" — for several distinct things, and from assuming that a method described as solving "task mining" solves the whole of it. This survey therefore organises the problem by *the assumption that the assignment withholds*, as summarised in Table 1 and Figure 2. Four such assumptions matter, and the literature that addresses each is largely disjoint.

Two consequences of this convention matter for the rest of the survey. First, a method that segments a stream and a method that labels segments solve different problems, and only a handful of works do both. Second, evidence about *boundaries* and evidence about *identity* are not interchangeable: a study reporting high task-switch detection accuracy has not shown that it can tell which process is being executed.

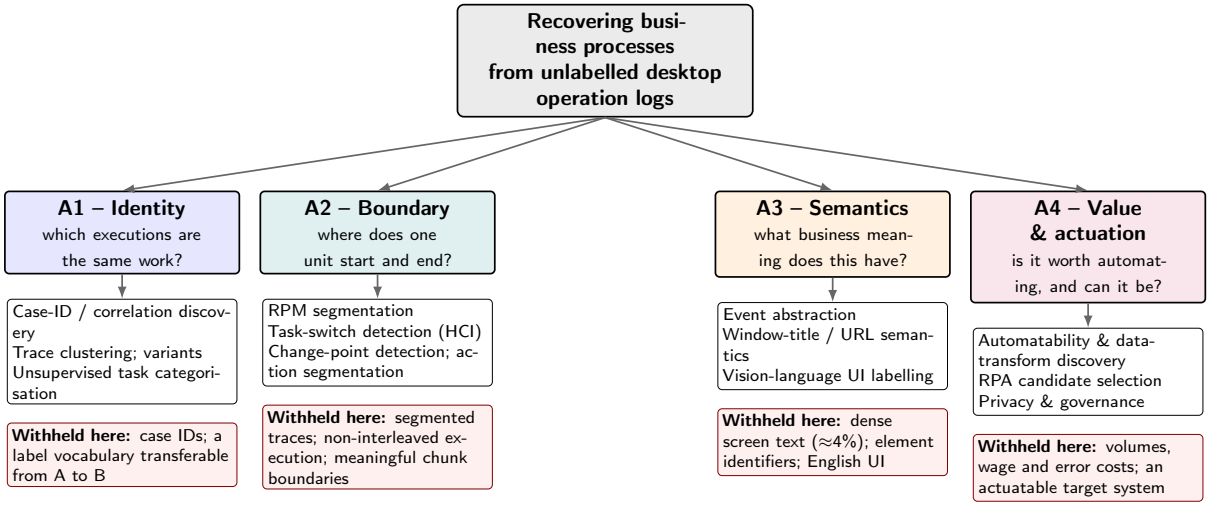


Figure 2: Taxonomy of the problem organised by the assumption the assignment withholds (A1–A4), with the research fields that address each and, in red, the specific thing this assignment removes. Original diagram prepared for this survey, synthesising the problem definitions in Leno et al. [31], Agostinelli et al. [4], van Zelst et al. [59] and Abb and Rehse [1].

3 Literature-Search Methodology

Two passes. The survey was conducted in two passes. The first established the field map and the principal near matches. The second was undertaken specifically to close gaps the first pass had itself flagged: sources blocked by publishers, Japanese-language work, segmentation-evaluation metrics, action segmentation, cross-domain transfer evidence, and the vendors whose documentation the first pass had not examined. The second pass changed four substantive conclusions, which are identified where they occur.

Venues and indexes searched. Springer Nature Link, IEEE Xplore, ACM Digital Library, ScienceDirect, arXiv, CEUR-WS, AIS eLibrary, dblp, Zenodo, HuggingFace, GitHub and GitLab; the IEEE Task Force on Process Mining resource pages; IEICE technical-report records; and vendor documentation sites. Within these, priority went to BPM and the BPM Forum, ICPM and its workshops, CAiSE, RCIS, ICSOC, ICIS, NeurIPS, CVPR, IJCAI, IUI, ACL, and the journals *BISE*, *Information Systems*, *ACM Computing Surveys* and *ACM TMIS*.

Table 1: Terminology used throughout this survey, organised by the assumption withheld.

Term	Assumption class	Definition adopted here
Event	—	One recorded low-level interaction: a keystroke, click, scroll, clipboard change, window-title change or browser action. The atomic unit of the raw log.
Chunk	—	A fixed recording-time bucket produced by the logging agent. A recording artifact with no business meaning; a unit of work may cross several.
Session	—	One continuous recording from start to stop of an employee's working session.
Segment / unit of work	A2 Boundary	A contiguous interval of a session judged to correspond to one execution of one business process. The object the assignment's `segments.jsonl` must contain.
Routine	A2 Boundary	The robotic-process-mining term for a repetitive, potentially automatable action sequence recovered from a UI log [31].
Process family vs. execution vs. task	A1 Identity	Three nested granularities present in this assignment's ground truth. A *family* is a kind of work; an *execution* is one instance of it, associated with a case; a *task* is a step inside an execution. Which of these constitutes "one unit" is an interpretation decision, not a given.
Variant	A1 Identity	A distinct handling pattern within one process family. Distinguishing a genuine business branch from operator habit, retry or error is not solved from logs alone.
Case identifier	A1 Identity	The attribute linking events belonging to one business case. Present by assumption in classical process mining; Leno et al. [31] state explicitly that UI logs do not have one.
Event abstraction	A3 Semantics	Mapping low-level events to higher-level business activities; a named sub-problem of process mining with its own taxonomy [59].
Segmentation	A2 Boundary	Determining which user actions contribute to which routine — identified by Agostinelli et al. [4] as the central open problem.
Interleaving	A2 Boundary	Concurrent execution of two or more processes, with the worker switching between them. Excluded by assumption in most published methods.
Shared actions	A2 Boundary	Actions that occur in more than one routine. The condition for which Agostinelli et al. [4] report finding no published solution.
Task mining	—	The commercial category for analysing desktop interaction data. Its products differ sharply in whether tasks are *discovered* or *configured* (Section 8).
Automatability	A4 Value	Whether a discovered routine can actually be executed by software. Leno et al. [31] note there is no formal definition and no guarantee that a frequent task is automatable.

Search terms. Combinations of: *robotic process mining*, *task mining*, *UI log segmentation*, *unsegmented UI logs*, *candidate routine identification*, *event abstraction*, *case ID discovery*, *correlation mining*, *task switch detection*, *desktop activity*, *window titles*, *trace clustering*, *process variants*, *concept drift*, *text segmentation metric*, *WindowDiff*, *temporal action segmentation*, *change point detection*, *vision-language UI labelling*, *RPA candidate selection*, *automatability*; and Japanese equivalents, transliterated here so that this document compiles without a CJK typesetting package — *sousa rogu* (operation log), *gyoumu purosesu* (business process), *tasuku maningu* (task mining), *gyoumu kashika* (work visualisation) and *jidouka kouho* (automation candidate). The terms in their original script, for direct re-use in a further search pass, are listed in the accompanying file `japanese-search-terms.txt`.

Date window. 2019–2026 primarily, with earlier work included where it is foundational or where it is the only evidence on real long-horizon desktop data — namely Oliver et al. [40], Shen et al. [50], Song et al. [52], Pevzner and Hearst [41] and Pourmirza et al. [42].

Inclusion criteria. A source was included if it (i) defines a method, benchmark or dataset for recovering process structure from interaction data, (ii) reports quantitative evaluation, (iii) is an authoritative statement of what the field considers open, or (iv) is official product documentation whose technical content is checkable. Preference was given to work with public code or data.

Exclusion criteria. Excluded: listicles and vendor comparison blogs; secondary summaries used as evidence for technical claims; and marketing pages used to support any claim about capability. Vendor marketing is reported in Section 8 but labelled as such and never used as evidence.

Verification protocol. Every source carries an explicit verification mark: • full text or substantial content retrieved and read during the survey; ○ abstract or landing page retrieved only; △ existence, title and venue confirmed through indexes but contents not retrieved. No △ source is used to support a central claim. Sources that resisted retrieval are listed in Section 11, together with author lists that the retrieved records left incomplete.

Limitations of the search. This is not a PRISMA-style systematic review: searches were conducted through general and academic web search without recorded result counts, so under-representation of very recent or poorly indexed work is possible. Three highly relevant publications could not be retrieved in either pass because of publisher access controls. Japanese-language academic coverage is materially incomplete: the CiNii search interface is robots-disallowed to the tooling used, and IPSJ special-interest-group reports are largely paywalled and weakly indexed in English, so the two Japanese academic items found (58, NTT; 53, Fujitsu) almost certainly under-count the domestic record. Negative findings in this survey are therefore claims about the reach of a bounded search, not proofs of absence, and are worded that way throughout.

4 Research Taxonomy and State of the Art

4.1 A1 — The identity boundary: which executions are the same work?

Classical process mining begins where this assignment ends. Its algorithms assume a *case identifier* linking events to a business case and an *activity label* naming each event, and given those, discovery, conformance checking and performance analysis are mature and well-tooled [15]. The literature that relaxes the case identifier is a recognised niche: correlation mining infers process orchestrations without case identifiers by exploiting statistical relationships between activity pairs and duration distributions [42], and case-notion discovery treats the case notion itself as a

choice to be searched over rather than a given [8]. Both assume attribute-rich structured logs; desktop telemetry offers almost no such attributes.

Identity across repeated executions, once segments exist, is the province of trace clustering [52] and variant discovery. De Weerdt et al. [19] show that the clustering choice materially changes the discovered model — relevant here because any segmentation and labelling decision propagates directly into the frequency and effort figures that a downstream operational analysis would report.

The one line of work that does identity *without* a label vocabulary is unsupervised task recognition. Rebmann and van der Aa [45] cluster recovered segments into task types using density-based online clustering, requiring no predefined number of types, and explain each cluster with tf–idf terms. Their evaluation reports clustering quality (Rand, Jaccard) separately from boundary quality — a two-part structure directly analogous to the assignment's requirement that boundaries be correct *and* that the same process receive the same label. Figure 3 shows the architecture.

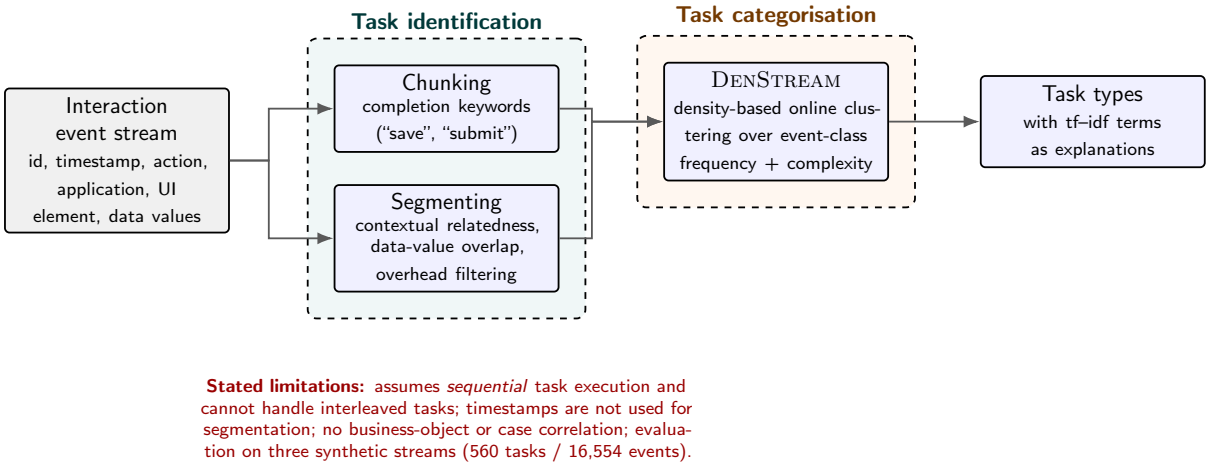


Figure 3: Unsupervised task recognition from an interaction stream: task identification followed by task categorisation. Redrawn by the author from the method description in Rebmann and van der Aa [45]; this is an original redrawing, not a reproduction of a published figure. The red note reproduces the authors' own stated limitations, of which the exclusion of interleaved tasks is the one that matters most for this assignment.

4.2 A2 — The boundary: where does one unit of work start and end?

This is the field's acknowledged hard problem and the assignment's Step 1. Leno et al. [31] name segmentation as a distinct pipeline stage and observe that UI logs carry no case identifier, so events cannot be correlated across applications by any existing key, and that purely time-based heuristics are unreliable because users are involved in different activities while performing a task.

Agostinelli et al. [4] surveyed the segmentation problem and classified its difficulty into three cases: repeated executions of a single routine; several routines without shared actions; and several routines *with* shared actions. Their conclusion is the sharpest negative result in this survey: only a few works can untangle unsegmented logs containing many interleaved routine executions, and they "did not find any literature work able to properly deal with user actions potentially shared by many routine executions." Figure 4 redraws that taxonomy and adds the condition this assignment describes.

Four method families attack the boundary problem. *Graph-and-pattern* approaches build a control-flow graph over the normalised log and derive boundaries from its cyclic structure before mining routines with sequential pattern mining [28]. *Semantic-cue* approaches chunk on interface completion keywords and contextual relatedness [45]. *Distributional* approaches detect shifts in

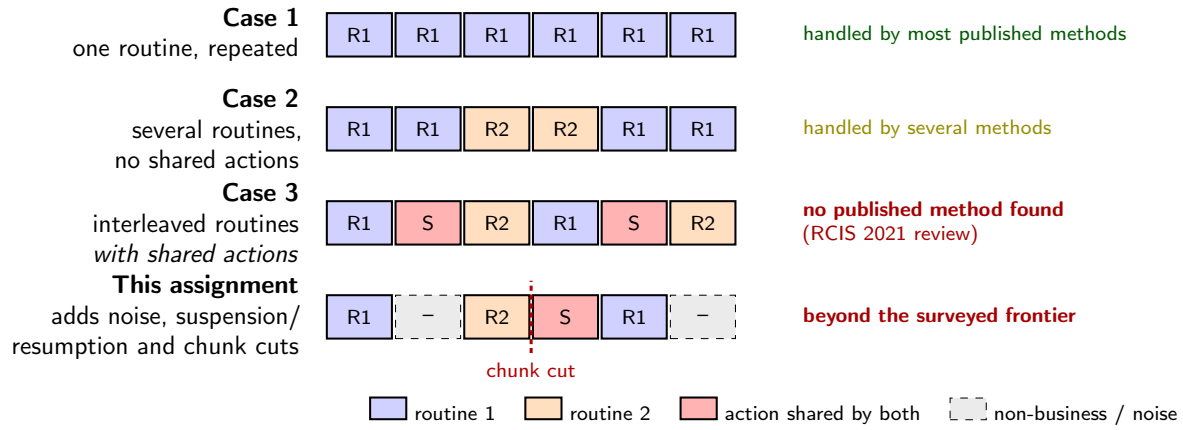


Figure 4: The three segmentation difficulty cases, with a fourth row showing the condition this assignment's brief describes. Rows one to three are redrawn by the author from the case taxonomy described in Agostinelli et al. [4]; the fourth row is this survey's own addition, derived from the assignment's data specification. Original redrawing, not a reproduction of a published figure.

the statistical composition of operations, borrowing directly from topic segmentation in natural language [58]. *Signal-processing* approaches encode the stream and apply time-series motif discovery [25], drawing on the generic machinery of change-point detection [55] and, in adjacent domains, on activity segmentation from sensor streams [7] and temporal action segmentation in video [22, 32]. A fifth response is to abandon full automation: Agostinelli et al. [3] propose frequent-pattern discovery followed by expert filtering, which is itself evidence about the state of the art.

Figure 5 redraws the most complete published attempt, and annotates the failure conditions its authors report. This is the finding that most changed between the two research passes: read only as an abstract, the paper appears to solve the assignment's Step 1; read in full, its stated preconditions are the negation of the assignment's data description.

4.3 A3 — Semantics: what business meaning does an action carry?

Three regimes of evidence exist, and this assignment falls between them.

Metadata only. Oliver et al. [40] showed that window titles alone support task grouping at roughly 70 % accuracy, rising to about 80 % when window-switching structure is added, on four hours of single-user labelled data. Shen et al. [50] used words from window titles and file pathnames, selected by information gain, and reached about 0.78 segmentation agreement on four real users recorded over two to six months — the only long-horizon real-desktop evaluation this survey found. Urabe et al. [58] built operation identity from application name, window title and GUI component information, with no screen text at all. Leno et al. [28] explicitly *discard* data values and keep only context parameters. The consistent message is that metadata alone supports meaningful but imperfect recovery, degrading as interleaving and application switching increase.

Dense screen capture. Screenshot-based task mining assumes every screen is captured and processed [33], and the emerging vision-language line takes this further. Rodríguez-Ruíz et al. [47] evaluate vision-language and large language models labelling UI-log activities from screenshots in settings where UI metadata is deliberately unavailable, such as virtualised desktops and business process outsourcing, reporting activity-level descriptions at 0.75 or better cosine similarity to human annotations, with hallucination, formatting instability and prompt sensitivity as named failure modes. Related work reconstructs UI hierarchies from logs, and general semantic enrichment of event-log text has its own line [11, 14, 17, 44].

The intermediate regime — roughly four per cent of events carrying screen text, plus selec-

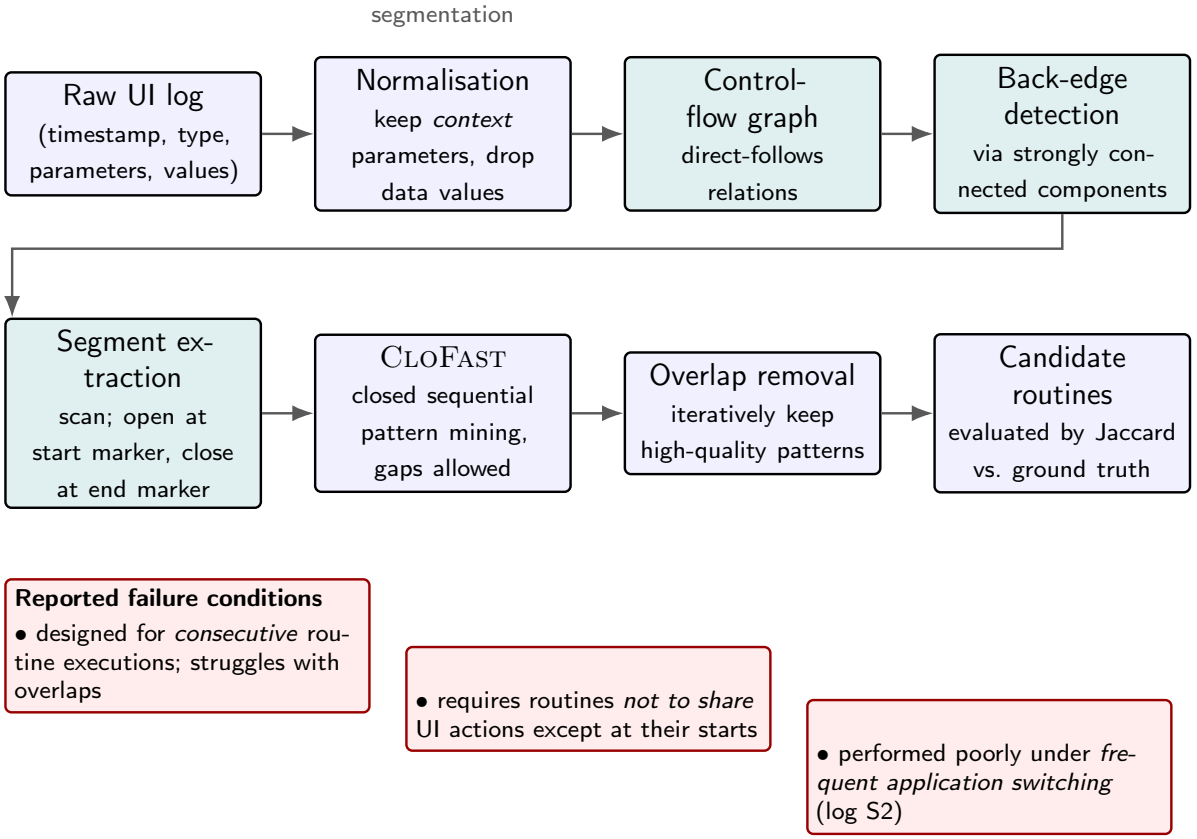


Figure 5: Discovery of candidate routines from an unsegmented, noisy UI log. Redrawn by the author from the method description in Leno et al. [28]; original redrawing, not a reproduction of a published figure. The three red boxes paraphrase the failure conditions the authors state in their own limitations discussion. All three describe properties this assignment's brief attributes to its data.

tive screenshots — is not evaluated anywhere this survey could find. Whether that sparse text materially improves on window titles alone is an open empirical question, not a settled one.

Compounding this, the semantic signal that does exist here is Japanese: business-process names, extracted screen text, UI labels, window titles and portal DOM attributes. The peer-reviewed record on Japanese office UI telemetry is very thin — two items across two passes [53, 58] — against a visibly active domestic commercial market (Section 8).

4.4 A4 — Value and actuation: is it worth automating, and can it be?

Leno et al. [31] state plainly that automatability lacks a formal definition and that there is no guarantee a frequent task is automatable; identifying non-deterministic actions, where a human supplies external input or judgement, is listed as unsolved. The concrete operationalisation of a *testable* determinism claim is narrow: Bosco et al. [16] treat automatability as a property to be tested on a discovered routine, and Leno et al. [30] target the specific case of routines transferring data between spreadsheets and web forms, applying data-transformation discovery with optimisations that exploit UI-log structure and the observation that cross-application transfers typically copy alphabetic and numeric tokens separately.

Selection and prioritisation form a separate, largely management-science literature. Wanner et al. [60] propose criteria beyond frequency; Author list unverified [9] and Promitzer et al. [43] continue that line. The most complete published *structure* is recent: Gomez and Castillo [24] chain a process taxonomy, a multi-criteria suitability index computed by analytic hierarchy process over transaction volume, rule-based standardisation, digital data availability, error and compliance risk, and process stability, a tool-tier selection step, and an ROI module producing labour savings, error-cost avoidance, payback and three-year net present value. Its authors state explicitly that it is a conceptual synthesis on synthetic, literature-informed data rather than an instrument calibrated on primary survey data, and validate only sensitivity to weight perturbation. That combination — a complete structure with a candid disclaimer — is instructive for how such an argument should be presented.

At the actuation end, synthesis of executable robots from UI logs is demonstrated [5, 6, 29], but always assuming a compatible recorder and a target application that can actually be driven. Finally, governance is not optional in this setting: Elkoumy et al. [21] give a threat taxonomy for process mining, and the concrete obligations are visible in vendor documentation (Section 8).

5 Consolidated Literature Table

Table 2 lists the sources most directly relevant to this assignment. The full matrix, with method, data, metrics and stated limitations for each, appears in Section A. Tier 1 denotes work whose method or findings bear directly on Steps 1–3; Tier 2 provides vocabulary, context or adjacent machinery.

6 Comparative Analysis

Every mature method assumes something the assignment removes. Classical process mining assumes a case identifier and an activity alphabet. Trace clustering assumes segmented traces. Supervised abstraction assumes a label vocabulary drawn from the same domain [54]. Script synthesis assumes an actuatable target. Screenshot-based labelling assumes dense capture. And two of the closest methods assume non-interleaved execution outright [28, 45]. The assignment removes all of these simultaneously, which is what makes it unusual rather than merely difficult.

Table 2: Core sources by category, with verification status and relevance. Verification: ● full text read; ○ abstract only; △ indexed but not retrieved.

T	Work	Venue / year	Ver.	Relevance to this assignment
Problem definition and open problems				
1	Robotic Process Mining [31]	BISE 2021	●	Defines the pipeline this assignment reproduces; states UI logs have no case identifier and that frequency does not imply automatability.
1	Challenge of automated segmentation [4]	RCIS 2021	●	Three-case difficulty taxonomy; reports that no published work handles actions shared across concurrent routines.
2	Reference data model for UI logs [1]	BPM 2022	○	Documents that every tool conceptualises UI logs differently, which is why bespoke schemas are normal.
2	UI logs: state of the art [2]	Inf. Syst. 2024	△	Likely the fullest inventory of UI-log formats and datasets; not retrievable in either pass.
Step 1: segmentation and consistent labelling				
1	Candidate routines from unsegmented UI logs [28]	ICPM 2020	●	The most complete attempt at Step 1. Its stated failure conditions — overlaps, shared actions, frequent app switching — describe this assignment's data.
1	Unsupervised task recognition [45]	CAiSE 2023	●	Closest method doing both boundaries and consistent typing without labels; explicitly cannot handle interleaved tasks.
1	Task clustering for RPA planning [58]	BPM 2021	●	Closest to Steps 1 and 2 together, built on exactly the available fields, reporting workload distribution. Japanese industrial group.
2	Interactive segmentation [3]	ICSOC 2021	○	The field's practical fallback: expert filtering of machine-proposed segments.
2	Time-series segmentation [25]	BPM 2025	○	Recent motif-based boundary detection released with code and data.
2	Noise-tolerant routine logs [10]	preprint 2025	○	Targets execution inconsistency directly; attribution unverified.
2	Two-level process models [13]	CAiSE 2024	○	Addresses the family/execution/task granularity question directly.
Boundary evaluation — from adjacent fields				
1	P_k critique and WindowDiff [41]	Comput. Ling. 2002	●	Five documented defects of the obvious segmentation metric; determines whether over- or under-segmentation is the safer error.
2	Boundary edit distance / `segeval` [23]	ACL 2013	○	Near-miss-aware scoring and, uniquely, inter-annotator agreement for segmentation.
2	MS-TCN / MS-TCN++ [22, 32]	CVPR 2019 / TPAMI 2020	○	Segmental F1 at IoU thresholds: the closest existing convention for scoring interval outputs.
Desktop activity: the realistic lineage				
1	Real-time task-switch detection [50]	IJCAI 2007	●	The only long-horizon real-desktop evaluation with genuine interruptions; ≈0.78 agreement on a simpler problem.
2	SWISH [40]	IUI 2006	●	Window titles alone carry substantial task signal; temporal structure adds to it.
2	Tasks and PC operation logs [53]	IEICE 2021	○	Japanese corporate research; links known tasks to logs — the reverse inference direction.
Semantics				
1	Screenshot-based activity labelling with VLMs [47]	BISE 2026	●	Demonstrates activity semantics recoverable from screen images without UI metadata; assumes a screenshot per event.
2	Drivers behind variable human actions [33]	Inf. Syst. 2023	△	Directly targets the variant-driver question; blocked in both passes.
2	Event abstraction taxonomy [59]	Granular Comp. 2021	●	The formal vocabulary for "raw events to units of work".
Step 2–3: value, automatability, actuation				
1	Data transformations for RPA [30]	preprint 2020	○	What a testable determinism claim looks like, and how narrow it is.
1	Data-driven RPA opportunity prioritisation [24]	preprint 2026	●	Most complete published structure for taxonomy → suitability index → tool tier → ROI; authors disclaim calibration.
2	Process selection in RPA projects [60]	ICIS 2019	○	Criteria beyond frequency.
2	SmartRPA [5, 6]	Comp. Ind. 2022 / SoftwareX 2025	○	The most complete published path from UI log to running robot.
2	Privacy in process mining [21]	ACM TMIS 2021	○	Threat taxonomy for telemetry containing clipboard and screen content.

Interleaving is the confirmed weak point, and three independent lines say so. First, the RCIS review found no published work handling actions shared across concurrent routine executions [4]. Second, the strongest recent unsupervised method excludes interleaving by assumption [45]. Third — and this is the second pass's principal finding — the flagship discovery method reports in its own limitations that it is designed for consecutive routine executions, struggles with overlaps, requires routines not to share UI actions except at their starts, and performed poorly on the one real log characterised by frequent application switching [28]. The only literature that engaged seriously with real interruptions is the 2006–2009 desktop lineage, which detected *switches* without identifying *processes* [20, 40, 50].

Reported accuracies are not comparable, because there is no shared metric. Four representative works use four conventions: normalised Levenshtein edit distance over segment sequences plus Jaccard over routine sets [28]; normalised edit distance plus Rand and Jaccard [45]; F-score on segment assignment [58]; and a segmentation-agreement score [50]. None is parameterised by boundary tolerance. Table 3 sets these against the metric conventions of neighbouring fields, which have already worked out the traps.

Table 3: Segmentation evaluation conventions. The lower block is standard in adjacent fields and, as far as this survey could determine, has never been applied to UI logs.

Convention	Used by	Property
Used in task mining — four papers, four conventions		
Normalised Levenshtein edit distance over segment sequences	Leno et al. [28]; Rebmann and van der Aa [45]	Compares sequences of segments; not parameterised by how far a boundary may fall from its reference.
Jaccard coefficient over discovered routine sets	Leno et al. [28]	Scores *which* routines were found, not where their boundaries fell.
Rand / Jaccard index over task-type clusters	Rebmann and van der Aa [45]	Scores identity assignment; says nothing about boundaries.
F-score on segment assignment	Urabe et al. [58]	Treats assignment as classification; sensitive to segment-length imbalance.
Segmentation agreement (COAP)	Shen et al. [50]	Designed for switch detection rather than interval recovery.
Standard elsewhere — not imported into task mining		
P_k	Text/topic segmentation	Penalises a miss roughly twice as heavily as a false alarm; over-penalises near-misses; leniency varies with segment-size variance [41].
WindowDiff	Text/topic segmentation	Counts boundaries inside the probe window; penalises a near-miss at roughly $2e$ for offset e; stable across segment-size distributions [41].
Boundary edit distance / B measures	Fournier [23]	Models near-misses as explicit edit operations; also supports inter-annotator agreement on segmentation.
Segmental F1 at IoU thresholds	Temporal action segmentation [22, 32]	Matches predicted to reference *intervals* above an overlap threshold — the closest analogue to scoring `(start, end, label)` tuples.

Figure 6 illustrates why the choice is consequential rather than cosmetic: under a P_k-like metric a boundary placed slightly off is punished about as hard as an unrelated false positive, which changes whether over- or under-segmentation is the safer error.

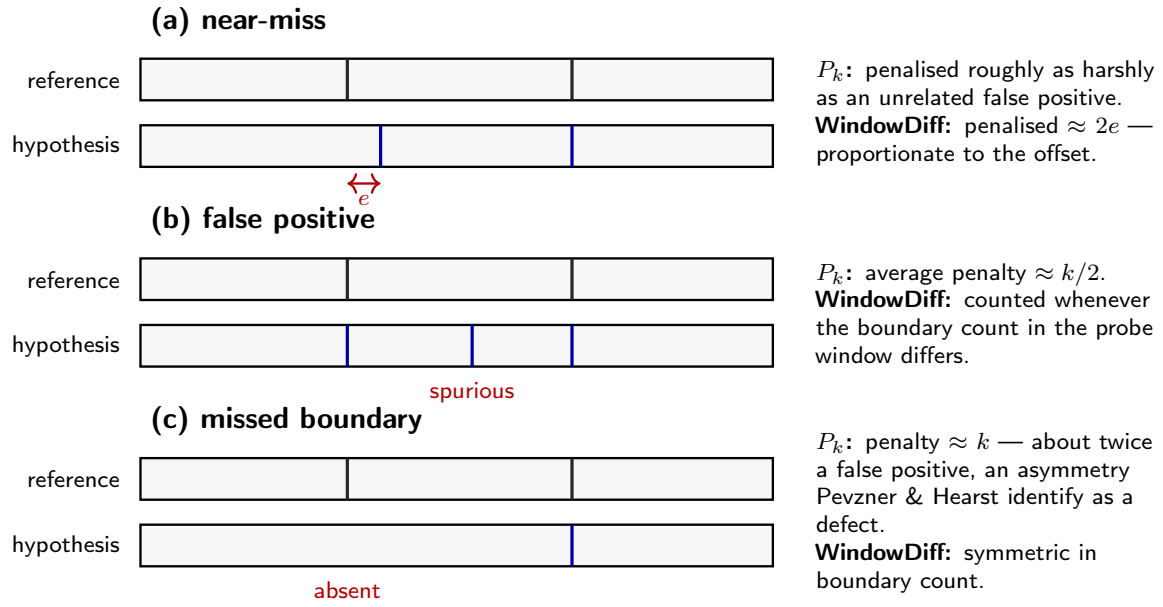


Figure 6: How the two standard linear segmentation metrics treat three error types. Original diagram prepared for this survey, illustrating the defects of P_k and the behaviour of WindowDiff as analysed by Pevzner and Hearst [41]. The figure is schematic: it depicts the qualitative penalty relationships the paper establishes, not measured values.

Coverage is uneven, and the gaps are structural rather than incidental. Figure 7 maps each subproblem of this assignment onto the field that would normally address it, distinguishing pairings routinely reported in the literature from pairings where the field exists but does not cover this project's condition.

Evaluations are small, and often synthetic. Published results in this area rest on evaluations one to three orders of magnitude smaller than this assignment's calibration dataset, and the largest are generated rather than recorded. Figure 8 plots the evaluation sizes that could be established from the papers themselves. Two caveats belong with it: Oliver et al. [40] and Shen et al. [50] report duration and user counts rather than event counts and are therefore omitted from the chart rather than misrepresented in it; and the scores those studies report are *not* mutually comparable, since as Table 3 shows they use different metrics. The figure is about the scale of the evidence, not about who performs best.

The validation asymmetry has no precedent in the literature. The assignment supplies ground truth for one department's processes and asks for delivery on another's. No task-mining study this survey found evaluates under that kind of distribution shift; the nearest adjacent literature is concept drift within a single log [49], which is a different phenomenon. Figure 9 states the resulting asymmetry, which governs how any accuracy claim in a downstream report must be worded.

Sparse semantics sits between two evidence regimes. Metadata-only methods report roughly 70–85 % on simple task sets and degrade with interleaving; dense-screenshot methods reach 0.75 or better activity-description similarity but presuppose a capture per event. The regime this assignment specifies — about four per cent of events carrying screen text, plus selective screenshots — is evaluated by neither.

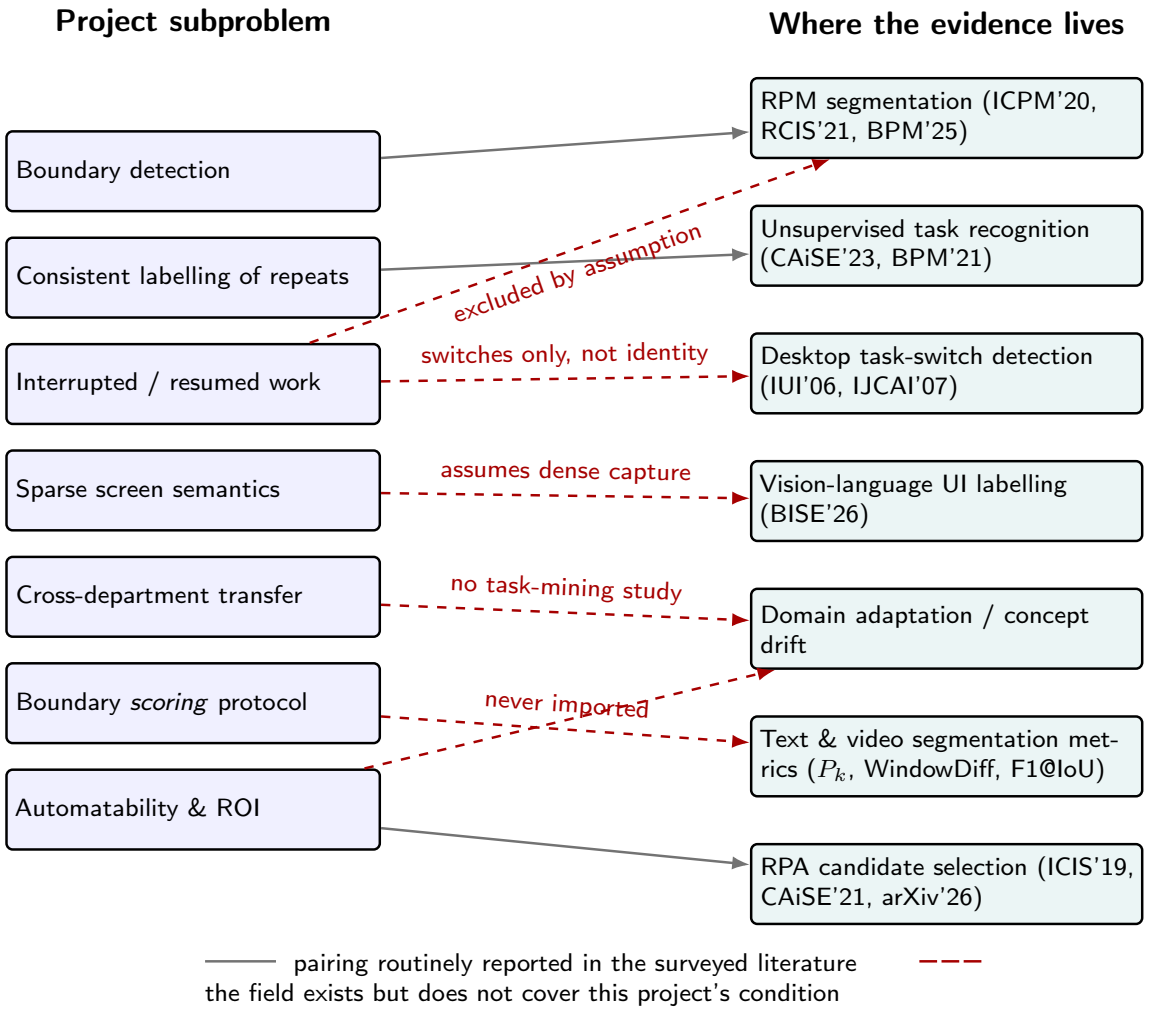


Figure 7: Coverage map relating this assignment's subproblems to the fields that address them. Solid arrows mark pairings routinely reported in the surveyed literature; dashed red arrows mark fields that exist but whose published results do not cover this project's condition. Original diagram prepared for this survey, synthesising Leno et al. [28], Agostinelli et al. [4], Rebmann and van der Aa [45], Shen et al. [50], Rodríguez-Ruíz et al. [47], Pevzner and Hearst [41] and Gomez and Castillo [24].

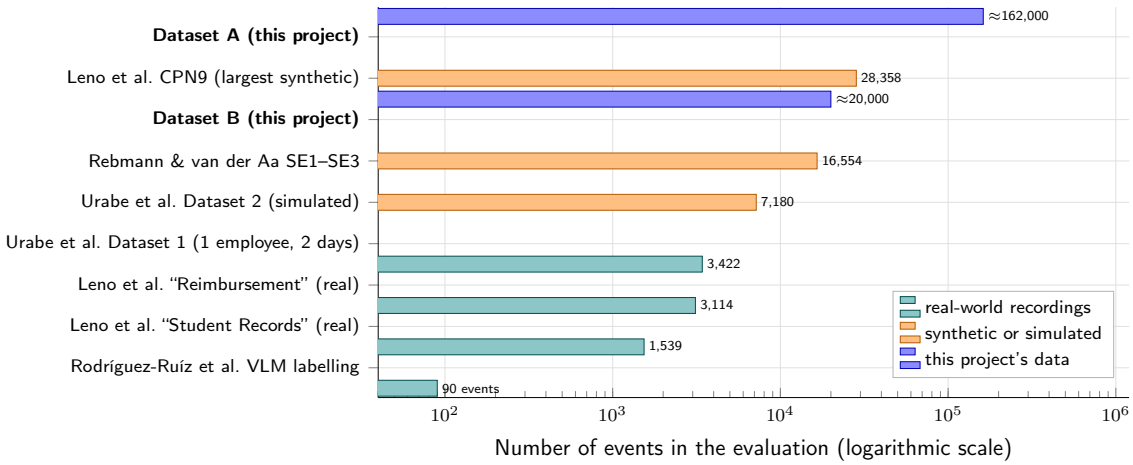


Figure 8: Scale of the evaluations underpinning the closest published methods, against the two datasets in this assignment. Original chart prepared for this survey from event counts reported in Rodríguez-Ruíz et al. [47], Leno et al. [28], Urabe et al. [58] and Rebmann and van der Aa [45]. Studies reporting duration rather than event counts (40: four hours, one user; 50: four users over two to six months) are omitted rather than approximated. Bar length indicates evidence scale only; the underlying accuracy figures are not mutually comparable.

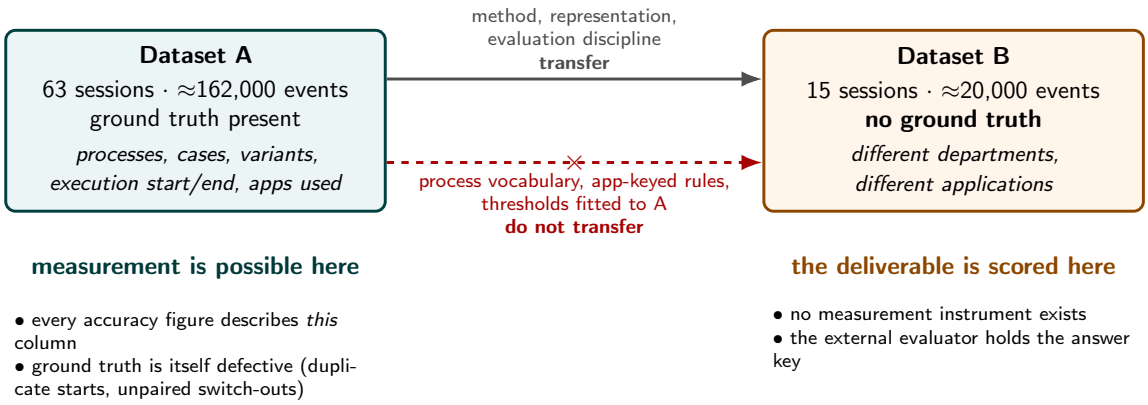


Figure 9: The validation asymmetry: measurement is possible only where the content does not matter, and the deliverable is scored only where measurement is impossible. Original diagram prepared for this survey from the assignment's own data specification.

7 Public Datasets, Benchmarks and Open-Source Landscape

7.1 Datasets and benchmarks

Two passes found no public dataset resembling this assignment's data. The closest UI-log datasets are small, single-process and screenshot-centric; the datasets containing rich desktop recordings come from the GUI-agent field and carry task labels by construction, which makes them useless as a *discovery* benchmark. Table 4 summarises.

Table 4: Public datasets and benchmarks, and why each is or is not a meaningful comparison.

Dataset	Modality	Access	Comparison value
UI logs from mockups / real screenshots [35]	CSV logs, PNG screenshots, BPMN and Petri-net models	CC-BY-4.0	The most directly comparable public UI-log dataset: one telecom unsubscription process, 10 activities, 1 decision point, 4 variants, logs of 75–500 events. Three orders of magnitude smaller and far cleaner, with dense screenshots rather than sparse text.
Annotated UI element dataset [34]	Annotated desktop screenshots, six application groups	CC-BY-4.0	Relevant only to screenshot understanding, not to segmentation.
Rebmann & van der Aa streams SE1–SE3 [45]	Synthetic interaction streams; 8 task types, 560 tasks, 16,554 events	via project repository	Closest analogue in *evaluation design* (boundaries and identity scored separately); synthetic and explicitly non-interleaved.
Leno et al. UI log corpus [28]	CPN1–9 synthetic (1,400–28,358 actions); Student Records and Reimbursement (real); Scholarship S1/S2	open via [25] repository	The de facto shared corpus of the sub-field, reused across many papers, so comparisons partly measure fit to one generator. S2 is the field's known hard case — and it is hard for the same reason this assignment is.
BPI Challenge logs [26]	Structured XES logs from real organisations	open	Case identifiers and activity labels are *present*. They begin where this assignment ends.
Process Discovery Contest [27]	Synthetic structured logs	open	Confirms that the community's competitive tradition operates entirely on labelled, case-ID-bearing data.
AgentNet / OpenCUA [62]	22.6K computer-use tasks; screenshots, actions, reasoning traces; Windows, macOS, Ubuntu	MIT	The richest public desktop-interaction corpus, but the inverse problem: identity is given and execution is the goal. Useful as evidence about what dense capture makes possible.
OSWorld [61]	Executable desktop environments with task specifications	open	Benchmarks agents *doing* work, not analysts inferring it.

7.2 Open-source software

Table 5 records what each relevant repository actually implements, verified by inspection. Two observations follow. First, no repository implements this assignment end to end: the mature tools sit on either side of the hard part, while the projects that address the hard part are single-lab research artifacts with single-digit adoption, one of them non-commercially licensed. Second, the one genuinely mature *evaluation* library comes from a different field and has apparently never been applied to UI logs.

Table 5: Open-source repositories, verified by inspection. Figures as displayed at the time of the survey.

Project	What it implements	Relevant to	Licence	Limitation for this assignment
PM4Py [15]	Classical process mining: log I/O, discovery, conformance, performance analysis	everything downstream of segmentation	open source (check current terms)	**Does not interpret raw UI or desktop logs.** No notion of keystrokes, window titles or segmentation from unlabelled telemetry.
ScreenRPA (RPA-US)	Screen-based task mining: UI recording, eye tracking, feature extraction, decision discovery, process visualisation	screenshot enrichment; decision points	CC BY-NC	Non-commercial licence; 6 stars; mostly notebooks; assumes screenshot-rich capture.
OpenAdapt	Records screenshots, GUI state, actions and accessibility metadata; conditions or fine-tunes vision-language models to replay tasks	the prototype/actuation end	MIT	Learns to *replay demonstrated tasks*; not a discovery tool for passive logs. Requires its own recorder.
TimeSeriesInRPM [25]	Motif-based UI-log segmentation (word2vec plus time-series motifs); bundles Leno-format data	boundary detection	MIT	Two notebooks, not a library; a reference implementation rather than infrastructure.
Task recognition from event stream [45]	Companion code for unsupervised segmentation and typing	boundaries *and* labels	not stated	Hosted on university GitLab; licence and maintenance unverified; implements the method that cannot handle interleaving.
SmartRPA [6]	Records UI logs, analyses variants, synthesises executable robots	actuation	see repository	Assumes its own recorder and an actuatable target.
`ruptures` [55]	Offline change-point detection: cost functions and search methods	generic boundary mathematics	BSD-2	Domain-agnostic; knows nothing about processes, identity or labels.
`segeval` [23]	P_k, WindowDiff, boundary edit distance, inter-annotator agreement	boundary *scoring*	see repository	Linear text-segmentation model; scores boundaries, not labels; would need adapting to timestamped intervals.
ActivityWatch	Cross-platform desktop time tracker: window and application watchers, AFK detection, rule-based categorisation	what desktop telemetry looks like in practice	MPL-2.0	Categorisation is **manual rule authoring**, not inference — a sanity check that this problem is normally solved by hand.

8 Industry and Commercial Landscape

Everything in this section comes from vendors' own documentation or marketing. None of it is independently validated. The useful structure is not "what do vendors claim" but "how checkable is the claim", which resolves the market into three groups (Table 6).

Governance is a shipped product requirement, and the obligation lands on the customer. UiPath's privacy documentation [56] is the most concrete material found: Task Mining records screenshots, keyboard actions, clicks and movements for configured applications; personally identifiable information is automatically masked; credit-card data is forbidden and protected health information requires a Business Associate Agreement; users may pause recording at any time and may view the full list of applications and interactions being recorded; data is encrypted in transit and at rest. Decisively, the documentation states that "the customer must take all the appropriate legal measures in order to ensure that data subjects are duly notified about the data capturing and processing initiative", because the vendor "has no technical possibility of identifying or taking consent from data subjects." The customer is the data controller, responsible for legal compliance, employee notification, access control and deletion. For a dataset containing clipboard contents, screen text and screenshots of employee work, that is the concrete shape of the governance obligation, and it complements the academic threat taxonomy of Elkoumy et al. [21].

The academic and commercial records disagree, but not uniformly. Papers say automatic segmentation of realistic interleaved logs is unsolved. The two best-documented products agree implicitly, by requiring a human to define the task first. The products that claim otherwise divide into those naming a mechanism and those naming only an outcome. That distribution — strongest claims where evidence is weakest — is itself a finding for any downstream risk assessment.

9 Research Limitations and Gaps

Table 7 distils sixteen limitations from the analysis above. Each is linked to its supporting evidence. Following the distinction the survey maintains throughout, a *broad open problem* ("machines cannot infer intent from actions") is not a gap because it is not testable; a *specific gap* names something currently unmeasured; and an *unresolvable question* is one that no reading can settle, deferred to Section 10.

9.1 Claims that would require especially careful evidence

Ranked by how easily an informed evaluator holding the answer key could challenge them.

C1. **Any accuracy claim about the unlabelled production dataset.** All measurement happens on the labelled dataset, which the specification declares to be a different distribution (L7). The nearest indirect evidence is discouraging: the one method evaluated across several real logs succeeded on some and failed on the one whose characteristics match this assignment's description [28].

C2. **Any absolute time-saving or ROI figure.** Waiting times are compressed by design; frequency does not imply automatability [31]; and the one complete quantitative framework found disclaims calibration [24]. Every step from a measured segment duration to an annual saving imports an assumption from outside the data (L14, L15).

C3. **Any claim that a discovered pattern is automatable.** Identifying non-deterministic actions — points where a human supplies external judgement — is listed as unsolved [31], and the testable version of the claim is narrow [30].

Table 6: Commercial landscape, grouped by how checkable the vendor's own account is.

Product	Stated input	What the documentation actually says
Group 1 — documentation admits human task definition		
Celonis Task Discovery [18]	Desktop activity via a client, plus business-application events	Users **manually create and define tasks**, specifying start and end events with up to three start and three end conditions expressed as event rules. A rule-based configuration system, not inference.
UiPath Task Mining [57]	Mouse clicks, keystrokes, hotkeys, across repeated executions of a *human-identified* task	Records and compares traces of a *known* task; AI clusters screenshots and merges traces into a task graph with decision points. Outputs a task graph, variant statistics, a Process Definition Document and a Studio XAML skeleton. States explicitly that it does **not** automatically discover unknown tasks and does **not** produce a ranked list of automation recommendations.
Microsoft Power Automate [36]	Desktop recordings of user actions	Pipeline: prepare recordings → analyse → visualise → identify recommendations → share. Outputs process maps, an application analytics report, and recommendations expressed as *which connectors to use*. No automatic grouping mechanism is documented. Notably, recommendations are framed by delivery feasibility rather than raw repetitiveness.
Group 2 — automatic discovery claimed in specific technical terms		
Automation Anywhere Process Discovery [12]	User interactions with various systems	Documentation states it captures, parses and analyses interactions; identifies and maps all relevant pathways *including variations*; automatically documents the most common workflow with minimal manual intervention. Outputs ROI comparison for prioritisation, a PDD, and a path to bot development. No method, accuracy or evaluation is published.
Nintex / Kryon Process Discovery [39]	**Computer vision** on employee desktops in real time, rather than backend systems	Claims visual process maps of user actions across all applications; metrics for employee count, frequency, duration, step count and variants; and automatic generation of a configured automation workflow exported to Kryon Studio. Marketing claims on the same page: up to 80 % reduction in RPA implementation time; a hotel-chain case study citing over $300k saved.
SAP Signavio Task Mining [48]	Application usage and switching, screen navigation, repetitive data entry, copy-paste, task sequences	States that AI is needed "because task-level data is often too detailed to review manually at scale", and is used to group similar actions, recognise patterns and highlight repetition. Grouping into "meaningful task sequences" is framed as analytical processing requiring interpretation, not push-button discovery.
Group 3 — outcomes claimed without checkable technical content		
Mimica [37]	Low-level desktop activity	Claims automated business process discovery "without interviews, surveys, or manual observation"; variation, decision points and exceptions down to screenshot level; a scoring engine ranking tasks by time savings, ease of automation and ROI; and PDD and BPMN outputs. **No method, evaluation or accuracy published.** These claims describe Steps 1–3 of this assignment almost exactly, which is why the absence of substantiation is worth recording.
Skan AI [51]	"Human-computer interactions across all applications"; "process metadata, not personal content"	Claims unsupervised continuous discovery and automatic identification of four "universal automation patterns": application switching, manual data re-entry, approval variability and exception handling. Quantified outcomes are marketing claims. The named patterns are an interesting hypothesis about what is detectable without business understanding, but they are asserted, not demonstrated.
Japan — the client's own market		
GenTaskMining [38]	PC operation logs	Launched 8 April 2026. A "Report Agent" gathers data from monitoring tools and creates documentation to help *identify business processes*; automation candidates are identified by comparing logged processes against actual workflow execution to surface differences between documented and actual procedure; an "Insight Agent" uses retrieval-augmented generation to provide improvement suggestions. The closest commercial analogue to this assignment found anywhere, in the right country, launched five months before the brief. Trade-press announcement only; no method public.

Table 7: Evidence-linked limitations in the surveyed literature, and the gap each implies.

#	Limitation	Evidence	Gap implied
L1	No published method handles actions shared across concurrent routine executions	[4]	The third and hardest of the three difficulty cases is unaddressed.
L2	The flagship discovery method requires consecutive, non-overlapping executions and degrades under frequent application switching	[28]	Its reported accuracies do not describe the conditions this assignment specifies.
L3	The strongest unsupervised method excludes interleaving by assumption and ignores timestamps for segmentation	[45]	No unsupervised method is validated on interleaved office work.
L4	Four papers use four incompatible evaluation conventions, none tolerance-parameterised	[28, 45, 50, 58]	Cross-paper accuracy comparison is unsound; a shared protocol does not exist.
L5	Mature tolerance-aware metrics exist in adjacent fields but have not been imported	[22, 23, 41]	The metric question is solved elsewhere and unasked here.
L6	Split (suspended and resumed) executions are treated as a detection obstacle, never as an output-representation problem	[4, 45]	How to *write down* a discontinuous execution in a flat schema is unstudied.
L7	No task-mining study evaluates under cross-department or cross-application distribution shift	survey-wide; nearest adjacent work [49]	How much accuracy survives such a shift is unknown, in either direction.
L8	Evaluations are small and frequently synthetic	Figure 8; [28, 45, 58]	Reported figures may partly measure fit to one generator or one operator.
L9	Real-data evaluations are one to two orders of magnitude smaller than this assignment's calibration set	[50, 58]	Scaling behaviour is unestablished.
L10	The intermediate semantic regime (sparse screen text plus selective screenshots) is unevaluated	[40, 47, 50]	Whether sparse text improves on window titles alone is an open empirical question.
L11	Vision-language labelling assumes a screenshot per labelled event and was evaluated on 100 screenshots in one process context	[47]	Selective-capture performance is unknown; hallucination and prompt sensitivity are named failure modes.
L12	Label granularity — family versus variant — has no published guidance, and interacts with scoring	[13]	An uncontrolled decision with direct consequences for any score.
L13	Distinguishing a genuine variant from habit, retry or error from logs alone is unsolved	[33] (not retrievable)	Variant claims cannot currently be grounded; the one paper targeting this is inaccessible.
L14	Automatability has no formal definition; frequency does not imply automatability	[16, 30, 31]	Testable determinism claims exist only for narrow data-transfer routines.
L15	Quantitative prioritisation frameworks are self-described as uncalibrated	[24]	ROI structures exist; empirically validated ROI instruments do not.
L16	Inter-annotator reliability of process ground truth is not studied in task mining	[23] treats it as first-class in text segmentation	Given that this assignment's ground truth is acknowledged defective, its own reliability is unquantified.

C4. **Any claim that a variant is a business rule rather than an artifact** (L13).
C5. **Any accuracy figure quoted without naming its metric and tolerance.** With four incompatible conventions in the literature and a hidden scorer, a bare number is close to uninterpretable, and an evaluator working in this field will know that (L4, L5).

10 Questions Only the Data Can Settle

The following cannot be resolved by further reading. They are recorded here so that a downstream analysis knows which of its assumptions are inherited from the literature and which must be established empirically.

Q1. What fraction of events carry any discriminative signal beyond timing and application context.
Q2. Whether window titles in the client's systems are informative — carrying case numbers, screen names or record identifiers — or generic. This single fact separates the regime in which Oliver et al. [40], Shen et al. [50] and Urabe et al. [58] reported usable accuracy from a materially harder one.
Q3. Whether the events carrying screen text cluster usefully (at process starts, at form submissions) or scatter at random.
Q4. How much interleaving actually occurs: what fraction of ground-truth executions are split by suspension and resumption, and how long the interruptions are. This determines whether the field's known failure mode (L1–L3) is actually triggered.
Q5. How often distinct processes *share* UI actions — the condition Agostinelli et al. [4] found unaddressed and Leno et al. [28] require to be absent.
Q6. How frequently the foreground application switches — the specific condition under which the flagship method degraded.
Q7. What the ground truth implies empirically about granularity: the ratio of families to executions to tasks, and the within-family variance (L12).
Q8. The boundary-time distribution: long idle gaps between executions, or immediate transitions.
Q9. The severity of the ground truth's own defects: the proportion of duplicated process-start records and unpaired switch-outs (L16).
Q10. Whether the production dataset's applications share any structural family with the calibration dataset's, or are genuinely disjoint (L7).
Q11. Whether the production sessions come from distinguishable individuals, and by which field.
Q12. Whether waiting-time compression is uniform across process types, which determines whether even *relative* duration comparisons are safe.
Q13. Whether clipboard contents or browser DOM attributes carry correlation keys — case numbers, employee identifiers — that could serve as a surrogate case identifier. This is the one route by which the correlation-mining literature [42] would become applicable.
Q14. What fraction of each session is non-business or unclassifiable activity.

11 Conclusion

This assignment is not a variant of a solved problem. Two research passes found no benchmark, competition, dataset or published method matching it, and the established competitive tradition in process mining operates entirely on structured logs that already carry the case identifiers and activity labels this assignment withholds.

Seven fields cover its components separately, and they are unevenly mature. Discovery from structured logs, trace clustering and segmentation *evaluation* are mature — the last of these in a neighbouring discipline that task mining has not drawn on. Event abstraction and case-identifier discovery are established but assume structured, attribute-rich input. Segmentation

of unsegmented, noisy, interleaved interaction logs is actively researched and explicitly unsolved, confirmed by three independent lines of evidence: a review that found no method handling shared actions; the strongest unsupervised method, which excludes interleaving by assumption; and the flagship discovery method, whose own limitations section names consecutive execution, non-shared actions and infrequent application switching as preconditions, and which degraded on the one real log that violated the last of them. Those three preconditions are the negation of this assignment's data description.

The commercial market tells a consistent story from a different direction. The two best-documented products require a human to define the task before analysis begins; the products claiming fully automatic discovery publish nothing checkable; and the vendor documentation that is most concrete is about privacy obligations rather than capability. A Japanese generative-AI service launched five months before this brief claims to do substantially what the assignment asks, and publishes no method at all.

What follows for a downstream analysis is not a technique but a discipline. Every accuracy figure will describe the calibration dataset, not the deliverable one. Every metric quoted without its tolerance is uninterpretable against a hidden scorer. Every ROI number imports assumptions the data does not contain. And every claim that a repeated pattern is automatable runs against an explicit statement, from the field that defined this problem, that frequency carries no such guarantee. The literature does not supply a solution to this assignment; what it supplies is an unusually clear account of which confident statements about it would be wrong.

Sources Requiring Further Verification

The following are recorded for completeness. None is used to support a central claim, and each should be checked against a publisher record before being cited in a formal submission.

- **Abb and Rehse [2]** — *Information Systems* (2024). Publisher blocks automated retrieval; no preprint located in either pass. Likely the fullest existing inventory of UI-log formats and public datasets, so its absence is the most consequential gap in Section 7.
- **Martínez-Rojas et al. [33]** — *Information Systems* 118:102340 (2023). Returned HTTP 403 and is robots-disallowed on both routes tried. It targets the variant-driver question directly (L13), which makes it the most consequential omission in Section 4. Title, venue and DOI were verified through two independent indexes.
- **Author list unverified [9]** — CAiSE 2021. Rate-limited on both attempts. Title and venue verified through Springer, the ACM Digital Library and the CAiSE 2021 accepted-papers list; **author list not established** and recorded as unverified in `references.bib`.
- **Author list unverified [10]** — arXiv:2510.08118. Content was readable and is summarised in Section A, but the arXiv pages retrieved did not return an author list or a venue. Attribution is therefore unverified.
- **Promitzer et al., *Objectively Assessing the Suitability of Digital Processes for Robotic Process Automation*** [43] — Wirtschaftsinformatik 2022. Indexed in dblp under the key `conf/wi/PromitzerTW22` but not retrieved; author given names not established.
- **Truncated author lists.** The records retrieved for Rodríguez-Ruíz et al. [47], Rehse et al. [46], Dietterich et al. [20], Burattin et al. [17], Xie et al. [61], Author list unverified [8] and XLANG Lab [62] did not give complete author lists; these entries use "and others" or an institutional author and must be completed before formal citation.
- **Japanese-language coverage.** The CiNii search interface is robots-disallowed to the tooling used, and IPSJ special-interest-group reports are largely paywalled. Only two Japanese academic items were located across both passes. A native-language pass through CiNii, J-STAGE and IPSJ SIG proceedings would likely change Section 4 and Section 9.

A Full Literature Matrix

Table 8: Full literature matrix. Verification: ● full text read; ○ abstract or landing page only; △ indexed but not retrieved. "Stated limitation" records what the authors themselves report; "added limitation" marks an issue identified by this survey's analysis.

T	Work	Venue, year	V	Problem addressed	Method	Data and evaluation	Findings	Limitations
1	Leno et al. [31]	BISE, 2021	●	Define robotic process mining and its open problems	Conceptual six-stage pipeline: recording, noise filtering, segmentation, simplification, candidate routine identification, executable routine discovery	None (position paper)	UI logs have no case identifier; time-based segmentation heuristics unreliable; automatability undefined; frequency ≠ automatable; non-deterministic action detection unsolved	*Added:* no method, no data, no transfer story; assumes a recorder capturing UI element identifiers
1	Leno et al. [28]	ICPM, 2020	●	Discover candidate routines from a single unsegmented, noisy UI log	Normalise log to context parameters; build control-flow graph of direct-follows; find back-edges via strongly connected components as segment start/end markers; sequential scan to extract segments; CloFast closed sequential pattern mining with gaps; overlap-removal optimisation	Synthetic CPN1–9 (100–2,000 segments; 1,400–28,358 actions); real supervised: Student Records (50 seg., 1,539 actions), Reimbursement (50 seg., 3,114 actions); real unsupervised: Scholarship S1, S2 and combined variants. Metrics: normalised Levenshtein edit distance; Jaccard for routine quality	Segmentation NLED 0.0 (synthetic), 0.059–0.331 (real); routine Jaccard ≥0.90 for most logs, coverage >0.90; runtimes 0.461–3.649 s	*Stated:* depends on data quality and routine frequency; sensitive to multiple routine start actions; spurious variants when order differs; **designed for consecutive executions, struggles with overlaps**; **poor under frequent application switching (log S2)**; **requires routines not to share actions except at starts**

T	Work	Venue, year	V	Problem addressed	Method	Data and evaluation	Findings	Limitations
1	Rebmann and van der Aa [45]	CAiSE, 2023	●	Segment and categorise tasks in an interaction stream with no labels	Chunking on interface completion keywords; segmenting via contextual relatedness (co-occurrence), data-value overlap and overhead filtering; DenStream density-based online clustering over event-class frequency and complexity; tf–idf explanations	Three synthetic streams SE1–SE3 combining 8 task types; 560 tasks, 16,554 events. Metrics: identified task count, normalised edit distance; macro/micro Rand, Jaccard; memory and latency	SE1 198/200 tasks at 0.04 edit distance, 0.95 Rand; SE2 231/240 at 0.06, 0.95; SE3 138/120 at 0.23, 0.88; memory <1 % of event volume; response 2–150 ms	*Stated:* evaluation data lacks realistic mixed process/overhead sequences; timestamps not used for segmentation; no case correlation; **assumes sequential execution, cannot handle interleaved tasks**
1	Urabe et al. [58]	BPM, 2021	●	Segment and cluster office tasks to plan RPA introduction	Segmentation by shifts in operation co-occurrence (analogy to topic segmentation); operation ID from application name + window title + GUI component; agglomerative hierarchical clustering on cosine similarity	Dataset 1: 1 employee, 2 days, 3,422 operations, 5 business tasks. Dataset 2: 7,180 simulated operations, 3 tasks. Metric: F-score	F = 0.84 (real), 0.93 (simulated); identifies task types and workload distributions	*Stated:* parameter optimisation necessary; segmentation fails when per-task operation counts vary significantly. *Added:* single-operator real evaluation; no interruption handling; no screenshots
1	Agostinelli et al. [4]	RCIS, 2021	●	What makes UI-log segmentation hard	Structured review; three-case difficulty taxonomy (single routine repeated; several routines without shared actions; several routines with shared actions); literature mapped onto the cases	Analytical, using a travel-authorisation example across Excel and web forms	Few works untangle interleaved executions; **no work found that handles actions shared across routine executions**; recommends moving from model-based to learning-based approaches	*Added:* provides no method and no quantitative comparison
2	Agostinelli et al. [3]	ICSOC, 2021	○	Segmentation with expert filtering	Frequent-pattern identification; human-in-the-loop rejection of invalid segments; trace alignment to cluster actions	Reported over segmentation variants	Human filtering compensates for false positives in automatic discovery	*Added:* not autonomous; evaluation detail unavailable in the retrieved record

T	Work	Venue, year	V	Problem addressed	Method	Data and evaluation	Findings	Limitations
2	Hohenadl [25]	BPM, 2025	○	Boundary-free segmentation of noisy logs	word2vec encoding plus time-series motif discovery	Code and data released; evaluation detail not in the retrieved record	Removes the need for predefined routine boundaries; targets high-frequency, high-similarity routines	*Added:* small artifact (0 stars, ∼20 commits); evaluation not verifiable from the landing page
2	Author list unverified [10]	preprint, 2025	○	Extract routine *logs*, not only routine actions, under noise	Clustering-based routine log extraction	Nine UI logs from prior literature with systematically injected noise at varying levels	Reports superiority over adapted state-of-the-art under noisy execution	*Added:* **author list and venue unverified**; reuses the same shared corpus as prior work
2	Barba et al. [13]	CAiSE, 2024	○	Hierarchical abstraction of UI logs	Screen-level activities identified alongside fine-grained actions; dual-layer representation	Synthetically generated UI event logs with realistic characteristics	Contextual information is essential to map low-level interactions to higher-level activities	*Added:* synthetic only; does not address label invention for an unseen domain
1	Shen et al. [50]	IJCAI, 2007	●	Detect desktop task switches in real time	Words from window titles and file pathnames, top-200 by information gain; single-window classifier combined over time by probability voting, likelihood-ratio testing, or Viterbi with task-transition costs	4 real users, 2–6 months each; 246–1,209 labelled switches; 8–176 distinct tasks; plus 20 synthetic datasets	Best segmentation agreement (COAP) ≈0.78; precision improved >15 % using transition costs	*Stated:* labels contain considerable noise; assumes at most one switch per observation window; detection delayed by queue size. *Added:* detects switches, not process identity
2	Oliver et al. [40]	IUI, 2006	●	Group windows into tasks	PLSI over tf–idf weighted, stemmed, compound-split window titles; window-switching transition matrix with clique detection; ensemble falling back to temporal evidence	4 hours, 1 user, 5 tasks, manually labelled	≈70 % accuracy from titles alone; ≈80 % adding temporal information	*Added:* very small evaluation; no process or case notion; 2006-era desktop environment
2	Tada et al. [53]	IEICE tech. report, 2021	○	Link task information to PC operation logs for worker self-awareness	Monitoring and visualisation presenting tasks together with the behavioural data recorded at that time	Evaluation of monitoring and visualisation	Helps workers understand their work situation and notice issues	*Added:* **presupposes task labels** and attaches logs to them — the reverse of the inference direction required here

T	Work	Venue, year	V	Problem addressed	Method	Data and evaluation	Findings	Limitations
1	Rodríguez-Ruíz et al. [47]	BISE, 2026	•	Semantic activity labelling of UI logs where UI metadata is unavailable (virtualised, BPO)	Vision-language and large language models prompted over screenshot crops; "Component Parent Crop" input with "Highlight" prompting	100 annotated desktop screenshots; 30 images → 90 events with ground-truth descriptions in 3 complexity groups. Models: InternVL2.5, Qwen2.5VL, NVILA, UI-TARS; GPT-4, Mistral-Large, Qwen2.5-72B, Llama-3.1-405B. Metric: cosine similarity (all-mpnet-base-v2)	InternVL2.5 best on events; activity-level similarity 0.75+ for best models; robust across component density	*Stated:* hallucination in generalist models; formatting inconsistency; prompt effectiveness varies by architecture. *Added:* one process context; assumes a screenshot per labelled event
2	Martínez-Rojas et al. [33]	Inf. Syst., 2023	△	Why the same task is executed differently	Screenshot-based feature extraction feeding decision discovery	Not retrieved	Not retrieved	*Added:* **full text blocked in both passes**; the most consequential omission of this survey
2	van Zelst et al. [59]	Granular Computing, 2021	•	Bridging low-level events and business activities	Survey and taxonomy of abstraction techniques: supervised, unsupervised, model-based	Literature analysis	Establishes event abstraction as a named, structured sub-problem with a taxonomy	*Added:* assumes case identifiers exist; nothing on desktop telemetry or interleaving
2	Tax et al. [54]	preprint, 2016	○	Abstract low-level events to high-level activities	Supervised sequence labelling	Labelled datasets	Abstraction is tractable when labels are available	*Added:* the labelled-same-domain assumption is exactly what the A/B split removes
2	Pourmirza et al. [42]	ICSOC, 2015	○	Discover process models with no case identifier	Statistical correlation between activity pairs plus duration distributions	Synthetic and real logs	Formalises "no case identifier" as a solvable problem class	*Added:* assumes a known activity alphabet and attribute-rich structured logs
2	Author list unverified [8]	BPI workshop, 2021	○	Discover the case notion itself	Attribute-driven search over candidate case notions	Event log experiments	The case notion is a modelling *choice*, not a given	*Added:* needs attribute-rich structured data; author list unverified

T	Work	Venue, year	V	Problem addressed	Method	Data and evaluation	Findings	Limitations
2	Song et al. [52]	BPM Workshops, 2008	○	Split heterogeneous logs into homogeneous groups	Profile-based feature vectors plus clustering	Real logs	Canonical "same process, different executions" machinery	*Added:* requires traces to exist — i.e. segmentation must already be solved
2	De Weerdt et al. [19]	IEEE TKDE, 2013	○	Clustering that improves discovered model quality	Model-driven active trace clustering	Real logs	The clustering choice changes the discovered model	*Added:* same structural assumption
1	Pevzner and Hearst [41]	Computational Linguistics, 2002	●	Evaluating linear segmentation	Analysis of P_k; introduction of WindowDiff, which counts boundaries inside the probe window	Simulation across error distributions and segment-size ranges	Five defects of P_k: asymmetric penalties (miss $\approx 2\times$ false alarm), missed errors in small segments, sensitivity to segment-size variance, over-penalisation of near-misses, poor interpretability. WindowDiff penalises a near-miss at $\approx 2e$ and is stable across segment sizes	*Added:* designed for text; would need adapting to timestamped intervals
2	Fournier [23]	ACL, 2013	○	Near-miss-aware segmentation evaluation	Boundary edit distance and the B family of measures; inter-annotator agreement for segmentation	Text segmentation corpora	Models near-misses as explicit edit operations; treats reference reliability as first-class	*Added:* linear text model; scores boundaries, not labels
2	Farha and Gall [22], Li et al. [32]	CVPR 2019 / TPAMI 2020	○	Temporal action segmentation in untrimmed video	Multi-stage dilated temporal convolutional networks	Standard action-segmentation benchmarks; metrics: frame accuracy, edit score, segmental F1@{10,25,50} IoU	Establishes the interval-matching evaluation convention	*Added:* fully supervised; different modality; imported here only for its metric
2	Truong et al. [55]	Signal Processing, 2020	○	Offline change-point detection	Cost functions × search methods (Pelt, binary segmentation, window, dynamic programming)	Library benchmarks	Generic mathematics of regime change	*Added:* domain-agnostic; no notion of process, identity or label
2	Aminikhanghahi and Cook [7]	PerCom workshops, 2017	○	Segment continuous sensor streams into activities	Change-point detection	Smart-home datasets	Unsupervised boundary detection on heterogeneous event streams is feasible	*Added:* sensor modality; simpler activity vocabulary

T	Work	Venue, year	V	Problem addressed	Method	Data and evaluation	Findings	Limitations
2	Sato et al. [49]	ACM Computing Surveys, 2021	○	Detecting behavioural change over time	Survey of drift detection and localisation	Literature	Vocabulary for distinguishing change from variation	*Added:* within-log temporal drift, not across-dataset domain shift
1	Leno et al. [30]	preprint, 2020	○	Identify routines transferring data between spreadsheets and forms	Data-transformation discovery applied to UI logs, with two optimisations exploiting log structure and separate alphabetic/numeric token copying	UI logs replicating a real-life repetitive data-transfer routine	Naive application of prior techniques is computationally inefficient; optimisations make it tractable	*Added:* the testable notion of determinism it supports is narrow
2	Bosco et al. [16]	BPM Forum, 2019	○	Detect deterministic, automatable routine fragments	Rule and data-transformation discovery over routine traces	UI logs	Introduces automatability as a property to be tested, not assumed	*Added:* assumes segmentation already done
1	Gomez and Castillo [24]	preprint, 2026	●	Identify and prioritise RPA opportunities	Four modules: process taxonomy (20 processes, 5 value streams); AHP-weighted Automation Suitability Index over volume, standardisation, digital data availability, error/compliance risk, stability; tool-tier selection; ROI quantification	Synthetic, literature-informed data; sensitivity analysis only (mean Spearman 0.83 over 2,000 Monte Carlo trials at ±20 % weight perturbation)	Most complete published structure for taxonomy → suitability → tool tier → ROI; ASI threshold 3.5 gates progression	*Stated:* "a conceptual synthesis. . . rather than an instrument calibrated on primary survey data". *Added:* sensitivity to weights is not correctness against outcomes
2	Wanner et al. [60]	ICIS, 2019	○	Quantifiable RPA candidate selection	Weighted decision criteria	Interviews and process documentation; expert validation	Criteria beyond frequency are necessary	*Added:* not log-driven
2	Author list unverified [9]	CAiSE, 2021	△	Identify and prioritise RPA candidates	Not retrieved	Not retrieved	Not retrieved	*Added:* **rate-limited in both passes**; author list unestablished
2	Agostinelli et al. [5, 6]	Comp. in Industry 2022 / SoftwareX 2025	○	Synthesise executable robots from UI logs	Reactive synthesis over log variants	Tool papers and case studies	The most complete published path from UI log to running robot	*Added:* requires a compatible recorder and an actuatable target application

T	Work	Venue, year	V	Problem addressed	Method	Data and evaluation	Findings	Limitations
2	Elkoumy et al. [21]	ACM TMIS, 2021	○	Privacy threats in process mining	Threat taxonomy and research agenda	Conceptual	Systematises confidentiality risk for event data	*Added:* not specific to UI logs, screenshots or clipboard content
2	Abb and Rehse [1]	BPM, 2022	○	What a UI log is	Reference data model implemented as an XES extension	Demonstrated in RPA contexts	Each study and tool uses a different conceptualisation, obstructing integration across sources	*Added:* not a mining method
2	Berti et al. [15]	Software Impacts, 2023	○	Process mining tooling	Library: log I/O, discovery, conformance, performance	Widely used	Mature open tooling for structured logs	*Added:* **does not interpret raw UI or desktop logs**

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